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This article introduces a framework to help researchers understand the mechanisms underlying product returns. The framework draws on research in consumer choice, consumer memory, and attitude stability to predict how the process that consumers go through at a predecisional stage affects their postpurchase behavior. The likelihood of product returns is considered contingent on the amount and nature of cognitive responses generated during the choice process. In Study 1, the authors focus on the impact of the nature of cognitive responses and show that the generation of responses of a different nature while choosing directly affects the likelihood of choice reversal. The comparative versus noncomparative nature of thoughts generated influences their diagnosticity and, accordingly, their impact on product returns when consumers are exposed to disconfirming information. In Study 2, the authors draw on the inoculation theory to manipulate the number of prechoice thoughts. They find that when faced with disconfirming information favoring a new brand, inoculated consumers who are presented with choice alternatives sequentially are less likely to return a brand than consumers who are exposed solely to positive information about the chosen brand at a prepurchase stage.

# The Impact of the Prechoice Process on Product Returns

Product returns cost retailers billions of dollars each year (Hess and Mayhew 1997). However, few researchers have focused their attention specifically on understanding the phenomenon of merchandise return. Much of the research that has been done on the topic has adopted a modeling perspective. For example, researchers have modeled the financial feasibility of return policies (Padmanabhan 1997) or a person's propensity to return a product based on demographic and behavioral databases (Hess and Mayhew 1997). Other research has focused on manipulating return policies to reduce product returns (Wood 2001). However, little attention has been directed toward investigating the process of returns—that is, what makes consumers reverse their decisions and return products, sometimes even before trying the product.

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The objective of this article is to develop a framework to improve the understanding of mechanisms that underlie consumer choice reversibility and thus consumers' likelihood to return products. The framework draws on research in consumer choice, memory, and attitude stability to predict how the process that consumers go through at a predecisional stage affects their postpurchase behavior. Studying the link between the prechoice process and the likelihood of product returns is important because (1) it improves the understanding of consumer behavior and (2) it provides managers with innovative strategies to reduce product returns by influencing what happens before choice.

## *SCOPE OF THE RESEARCH*

After consumers have chosen a product, they are often faced with information that implies that their choice might be inappropriate. For example, a consumer who orders a Sony CD player may hear from a friend that Aiwa CD players are better at playing scratched disks. In this scenario, the consumer is exposed to "disconfirming information," or the information from the buyer's environment (e.g., stores, friends, media) that implies that the choice of a particular alternative is inappropriate. In the preceding scenario, exposure to disconfirming information may result in the consumer's choice to return a product. In this article, we focus on such cases of product returns.

For many consumer products and services, the act of purchase and the act of consumption occur at different times. For example, time delays can be found in reserving a hotel room, ordering through catalogs or online, and buying durable products. In all of these contexts, there is a time delay between choice and consumption during which consumers may be exposed to disconfirming information that might, in turn, lead to the reversal of their initial choices. Thus, studying product returns as a result of exposure to disconfirming information has implications that apply to a wide array of consumer experiences and contexts.

Our decision to focus on product returns resulting from exposure to disconfirming information has several additional motives. First, product returns resulting from poor product performance tie in with research on customer satisfaction, an extensively researched topic. Second, despite the absence of detailed statistics on the various reasons for product returns, practitioners' literature implies that a substantial proportion of returns result from exposure to disconfirming information. Third, negative and comparative advertisements, one form of disconfirming information, are on the rise (Pfau 1997). Finally, and perhaps most important, previous research findings imply that in the case of reversible decisions, decision makers tend to scrutinize dissonant information, as predicted by utility theory (Frey and Rosch 1984), rather than avoid it (Frey 1981).

Despite its focus on the period directly following purchase, this research differs radically from research on cognitive dissonance. Specifically, a prerequisite for cognitive dissonance is the existence of social commitment (Korgaonkar and Moschis 1980). The absence of commitment (i.e., the ability to reverse a decision) directly implies a lack of cognitive dissonance. As such, according to theory, cognitive dissonance would not be present in the type of choice situations we study here in which the possibility of reversing the decision is a major part of the given. However, it is worth noting that some dissonance may arise when the return process is costly and aversive.

Throughout this article, we consider a scenario in which a consumer has made a choice between two brands, A and B, and has chosen A. Therefore, Brand A refers to the chosen brand, Brand B refers to the rejected brand, and Brand C refers to a brand that was not initially in the set of alternatives offered to the consumer.

## CONCEPTUAL MODEL

### *Informational Inputs to Return Decisions: The Accessibility–Diagnosticity Model*

Given the option to return a purchased product, consumers who face disconfirming information have, in effect, a "new" decision to make: Should they retain or return the chosen product? Because it is unlikely that consumers will go back and recheck all information about the chosen alternative, the decision to be made is a "mixed" one. A mixed decision is a task in which some information is physically present but other relevant information is stored in memory (Lynch and Srull 1982).

Similar to other mixed-choice situations, the information that consumers retrieve to help decide whether to return a product is a function of the accessibility of that information and its perceived diagnosticity (Feldman and Lynch 1988; Lynch, Marmostein, and Weigold 1988). The accessibility of information refers to the ease with which consumers can retrieve the information from memory, and diagnosticity is the degree to which the use of such information enables consumers to achieve their objectives in the particular decision task at hand (Lynch, Marmostein, and Weigold 1988).

A necessary condition for accessibility is availability. For consumers to be able to access thoughts from memory about the chosen alternative, they must have generated those thoughts. Thus, we argue that the greater the number of favorable cognitive responses developed while choosing, the greater is a person's ability to defend against attacks such as disconfirming information.[1] This ability may be the result of a larger storage of cognitive responses in favor of the chosen alternative (i.e., Brand A) that a person can draw on to support his or her choice. This argument parallels similar arguments that have emerged in the attitude literature in recent years (see, e.g., Haugtvedt et al. 1994).

### *The Nature of Prechoice Cognitive Responses: Comparative Versus Noncomparative*

Previous research has not directly addressed the relationship between the nature of cognitive responses generated while choosing and product returns. However, a thorough examination of research studies investigating different forms of processing (e.g., brand versus attribute processing) (see, e.g., Bettman 1982) and, particularly, studies with detailed analyses of verbal protocol data in choice processes (Bettman and Park 1980; Biehal and Chakravarti 1982) suggests the existence of cognitive responses of different natures. We propose that this difference has implications for choice reversibility.

Drawing on the coding schema that Bettman and Park (1980) develop and Biehal and Chakravarti (1982) subsequently apply and assess, we argue that the cognitive responses that consumers generate during the choice process have different levels (or degrees) of comparison among the considered alternatives. In effect, when comparing two alternatives, Brand A and Brand B, cognitive responses seem to fall into one of two broad types: (1) comparative and (2) noncomparative. This is in addition to the irrelevant or idiosyncratic thoughts that Wright (1973) defines. We define comparative thoughts as those for which consumers directly compare the given alternatives. Examples of comparative thoughts are "Because A has a longer warranty period than B, I will choose A," and "Why should I pay $15 more for A?" Noncomparative thoughts are broader in nature and do not make a direct comparison between the alternatives. Examples of noncomparative thoughts are "A is affordable," and "It is great that A has an excellent sound system."[2]

The diagnosticity and thus the use of prechoice cognitive responses in a return decision are contingent on both the

[1]Note that it is not always necessary to have greater numbers of stored cognitive responses for the chosen brand (i.e., Brand A). The stored responses that are accessible for subsequent use should have greater evaluative weight that favors Brand A.

[2]Noncomparative thoughts differ from overall evaluations. Noncomparative thoughts can be related to an attribute or can even imply a comparison, but they usually involve an evaluation that is broader in nature than a simple comparison between Brand A and Brand B. For example, the attribute can be analyzed in relation to monetary resources (i.e., this brand is affordable) or in relation to other brands on the market (i.e., this brand's sound system is one of the best).

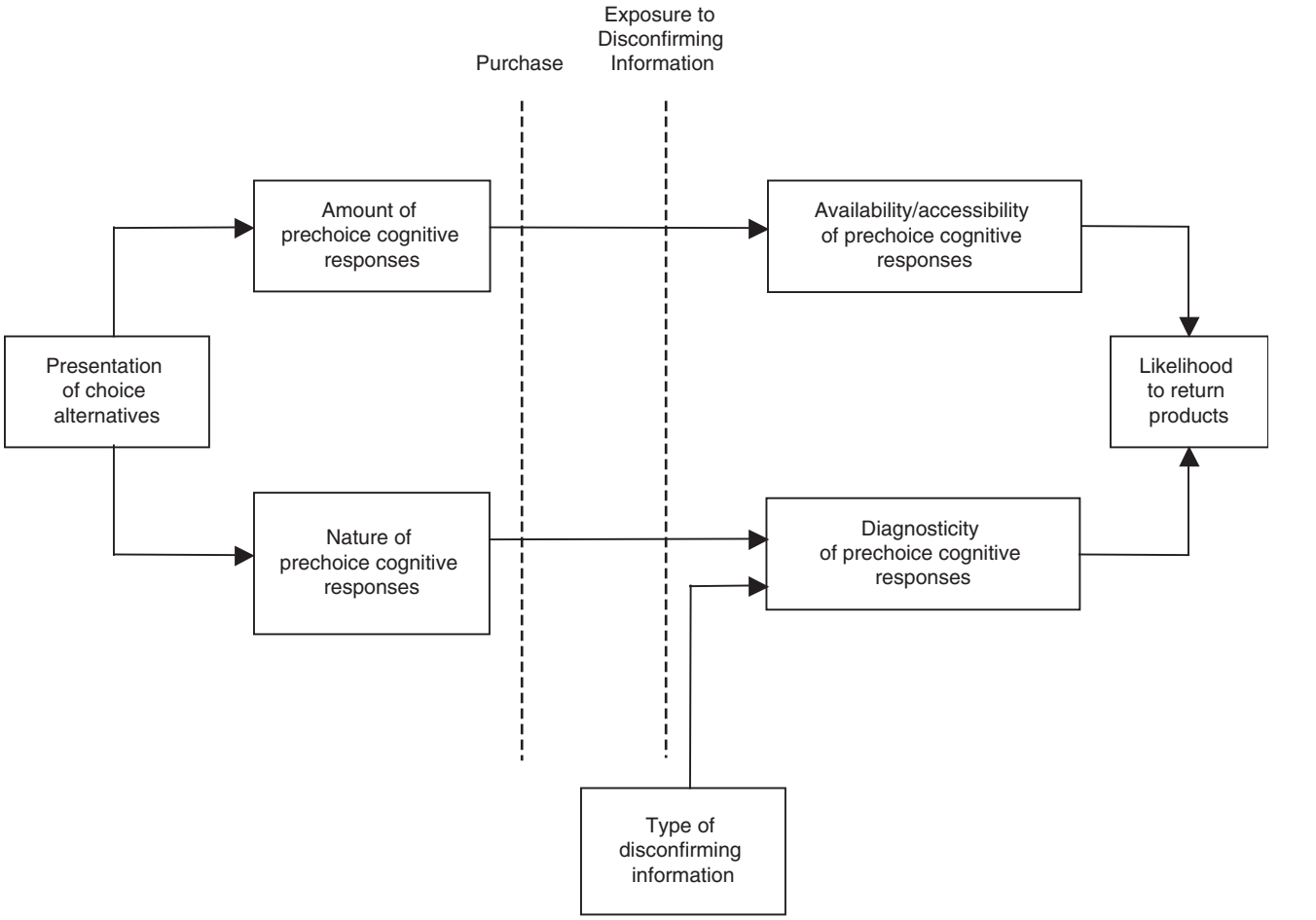
nature of the cognitive responses generated while choosing (i.e., comparative versus noncomparative) and the type of disconfirming information. For example, when disconfirming information favors Brand B, both comparative and noncomparative thoughts will be diagnostic. However, when disconfirming information favors Brand C, an alternative that was not initially considered in the choice decision, we expect that only noncomparative responses are diagnostic. Comparative responses developed before choice that praise Brand A over Brand B are of little use in defending Brand A against Brand C. We expect that only noncomparative responses praising Brand A are useful in defending a consumer's choice (Brand A) when disconfirming information favors Brand C.

In summary, we propose that the likelihood of returning a product after exposure to disconfirming information is linked to what happened before choice. This link is contingent on the accessibility and diagnosticity of prechoice thoughts at the time of the return decision. Three factors, among others, primarily determine the accessibility and diagnosticity of prechoice thoughts: (1) the number of favorable thoughts initially generated, (2) the nature of those thoughts, and (3) the nature of the disconfirming information. The first factor is derived from the accessibility or, more precisely, the availability principle. The other two factors determine the relevance of the prechoice thoughts (i.e., their diagnosticity). Figure 1 illustrates our conceptual model.

By focusing on mixed-choice situations, our research substantively resembles previous research on consumers' use of both memory and external information in choice. For example, Biehal and Chakravarti (1983, 1986) required participants to rethink a prior choice while giving them additional information in the form of more brands and changing choice criteria. Their results show that differences in the accessibility of prior information in memory significantly affect brand choice outcomes.

In our conceptualization, we adopt cognitive response approach to a choice issue. However, the adoption of the cognitive response approach—an approach supported in attitude research (see, e.g., Haugtvedt et al. 1994; Petty, Haugtvedt, and Smith 1995)—to choice contexts to predict the likelihood of product returns is not entirely straightforward. Considerable research suggests that attitude formation and choice tasks differ in decision processes (Alba, Hutchinson, and Lynch 1995). Thus, the extension of the cognitive response approach to the choice contexts that we propose is not straightforward and requires empirical verification.

Figure 1
THE IMPACT OF THE PRESENTATION OF CHOICE ALTERNATIVES ON PRODUCT RETURNS



## APPLICATIONS OF THE MODEL

We examine the application of the proposed link between prechoice process and postchoice product returns in two cases. In the first case, we manipulate the presentation format (simultaneous versus sequential) of choice alternatives to examine whether the generation of cognitive responses of different natures (comparative versus noncomparative) leads to a different likelihood of product return. In the second case, we draw on inoculation theory to identify contexts in which consumers generate different numbers of favorable thoughts.

In all cases, we assume that the level of decision makers' involvement is high. People are cognitive misers who are not expected to expend cognitive effort on unimportant issues (Burnkrant 1976). However, high involvement increases the likelihood that people acknowledge the vulnerability of their choices (Laurent and Kapferer 1985) and thus act to bolster them. We do not expect an uninvolved decision maker to care enough to generate a significant number of cognitive thoughts.

### *Presentation of Alternatives: Simultaneous Versus Sequential*

Although brand processing is not identical to the generation of noncomparative responses, we expect that it (versus attribute processing) results in a greater generation of noncomparative (versus comparative) thoughts. When consumers process by brand, they examine the brand along its various attributes before they examine another brand. This implies that more noncomparative thoughts will be generated. Dhar (1996) finds that brand processing is more likely to occur when information about alternatives is presented sequentially than when it is presented simultaneously. Accordingly, we expect decision makers who are presented with alternatives sequentially (versus simultaneously) to develop more noncomparative (versus comparative) thoughts about the chosen alternative (Brand A) before choosing.

The mode of presentation of choice alternatives affects the likelihood to return a product. When consumers are faced with disconfirming information favoring an alternative that was not in the initial choice set (Brand C), noncomparative thoughts, which we expect to be generated primarily when the presentation is sequential, will still be diagnostic. Therefore, noncomparative thoughts will be used in defending the chosen alternative (Brand A). Conversely, comparative thoughts that directly favor Brand A over Brand B, which we expect to be generated primarily when the presentation is simultaneous, will not be helpful in defending Brand A over Brand C. Accordingly, we posit the following:

$H_1$: In high-involvement choice situations, when consumers are exposed to disconfirming information favoring Brand C, those who are presented with choice alternatives sequentially are less likely to return products than those who are presented with choice alternatives simultaneously. However, when the disconfirming information favors Brand B, the return likelihood does not differ by whether choice alternatives are presented sequentially or simultaneously.

### *Inoculation Theory*

Inoculation theory posits that when people face an attack on an existing attitude, they will be more resistant to persuasion if they were previously exposed to a weakened (or mild) version of the countervailing argument (McGuire and Papageorgis 1961). This is analogous to immunization in biology. A relatively weak attack should encourage people (latently) to develop defenses against such attacks but not change their attitudes or beliefs (see, e.g., Kamins and Assael 1987).

Inoculating consumers might have an impact on product returns. Exposing decision makers to a low dose of negative information about Brand A at a prepurchase stage causes them to worry that they might not be making the appropriate choice. In turn, this threat motivates decision makers to develop reasons in favor of the chosen alternative, reasons that will enable them to resist disconfirming information when they are exposed to it later. Therefore, counterintuitively, we expect decision makers who are exposed to a low dose of negative information about a chosen alternative (Brand A) at a prepurchase stage to be less likely to return a purchased product than consumers who are exposed solely to positive information.

As we discussed, we expect that inoculation results in a decision maker bolstering his or her choice at a prepurchase stage. Some of the cognitive responses generated in the bolstering process should be noncomparative. When consumers are faced with disconfirming information favoring Brand C, we expect that such noncomparative thoughts in favor of Brand A are diagnostic when decision makers compare Brand A with Brand C. Thus, noncomparative thoughts will be used to defend the initial choice (Brand A). However, we expect those effects only when consumers are sufficiently involved to generate thoughts and when noncomparative thoughts are generated (i.e., when alternatives are presented sequentially). Thus, defining inoculation as exposure to a mild level of negative information about the chosen alternative at a prepurchase stage and on the basis of the preceding arguments, we posit the following:

$H_2$: In high-involvement choice situations, when inoculated consumers are exposed to disconfirming information favoring Brand C, those who are presented with choice alternatives sequentially are less likely to return Brand A than (a) consumers who are exposed solely to positive information about Brand A at that same prepurchase stage and (b) inoculated consumers who are presented with choice alternatives simultaneously.

## PILOT STUDIES

To identify contexts in which an adequate number of measurable prechoice thoughts are generated, we ran two pilot experiments. The first experiment consisted of a 3 (level of negative information about the chosen alternative [BrandA] at the prepurchase stage: none, mild, and high) × 2 (involvement: low and high) between-subjects design. We divided the experiment into two parts, which were separated by a period of two days. In the first phase of the study, decision makers chose between two alternative brands of a portable CD player. After two days, we exposed participants to disconfirming information. Disconfirming information took the form of a comparative advertisement that praised the rejected brand (Brand B) on an attribute that was not available in the chosen alternative (Brand A). We then gave participants the option to return the product they "purchased" in the first phase of the study.

We designed two measures to assess the dependent variable, likelihood of product returns: one expressed and the other behavioral. First, we measured the likelihood of product returns by checking whether participants indicated a final choice in Phase 2 of the experiment that was different from the one they initially indicated in Phase 1 of the experiment (i.e., measure of expressed reversibility). Second, we checked whether participants who expressed an intention to reverse their choice detached and deposited a return slip (as instructed) to activate their choice reversibility (i.e., behavioral measure).

We asked participants to list their thoughts in both phases of the study. Two judges independently evaluated the cognitive responses obtained through thought listing. The coding schema the judges applied in Phase 1 included three categories: (1) arguments in favor of Brand A, (2) arguments not in favor of Brand A, and (3) irrelevant thoughts. In assessing thought listings, we determined the number of net arguments in support of Brand A by the number of responses coded as "in favor" less the number of thoughts coded as "not in favor." Previous researchers have used the concept of net thoughts in coding cognitive responses (see, e.g., Wansink, Ray, and Batra 1994).

We analyzed data from 130 participants. In high-involvement choice situations, the percentage of inoculated participants who reversed their choice (13%) was lower than the percentage of participants who were exposed to no negative information at the prechoice stage (47%; $p < .01$). We found no such difference under low involvement (17% versus 20%; $p > .5$). In coding participants' thoughts in Phase 1 of the experiment, the level of agreement between the two judges was 87%. The mean number of net arguments in favor of Brand A averaged 2.10 across all experimental conditions and ranged between 1.68 (for low involvement and no negative information) and 3.05 (for high involvement and mild negative information).

The second pilot study had the same design as the first. The major difference between the two studies was that in the second experiment, the disconfirming information in the advertisement favored Brand C rather than Brand B. In high-involvement choice situations, the percentage of inoculated participants who reversed their choice (10%) was lower than the percentage of participants who were exposed to no negative information at the prechoice stage (40%; $p < .05$). We found no such difference under low involvement (2% versus 10%; $p > .1$). The level of agreement between the two judges was 91%. The mean number of net arguments in favor of Brand A in Phase 1 averaged 2.03 and ranged between 1.63 and 3.27 across all experimental conditions.

The pilot studies provided assurance that it was possible to measure thoughts generated by participants. The number of net arguments generated was not negligible. The coding schema was applicable without major problems. The level of agreement between the two judges was acceptable and reassuring. This finding was particularly true in high-involvement choice situations and held true whether disconfirming information favored the rejected brand (Brand B) or a new brand (Brand C). Thus, we conducted Studies 1 and 2 using high-involvement choice situations. Both pilot studies contained a coding task focused on the second choice phase that did not yield usable data because of a large number of irrelevant thoughts. To overcome this problem, we gave participants more specific instructions in Study 1. In Study 2, we measured the latency of thoughts to capture their accessibility or retrieval.

## *STUDY 1*

### *Method*

*Overview*. The experimental design was a 2 (presentation of choice set: sequential and simultaneous) × 2 (type of disconfirming information: favoring Brand B and favoring Brand C) between-subjects design. A total of 87 undergraduate students at a northeastern university participated in the experiment. We ran the experiment in two phases with a two-day delay. In Phase 1 of the study, participants chose between two equally attractive brands of a portable CD player (Brand A and Brand B). Exposure to disconfirming information occurred in Phase 2. Disconfirming information took the form of a comparative advertisement that praised either the rejected alternative (Brand B) or an alternative that was not initially in the choice set (Brand C).

*Procedure*. We ran Phase 1 of Study 1 in a similar manner to the pilot studies. We described alternative brands of portable CD players in a brand-by-attribute matrix on six attributes. These attributes were ease of use, sound accuracy, availability of a low-battery indicator, average alkaline battery life, handling of problem disks, and availability of an automatic power-off. Pretesting (n = 62) revealed equal preference for the two alternatives (Brand A and Brand B).

We presented descriptions of the two brands either simultaneously (i.e., on the same page) or sequentially (i.e., on two consecutive pages). We randomly mixed booklets before the experiment to ensure a random assignment of the participants. We rotated the order of the choice alternatives (Brand A or Brand B). After making a choice, participants listed the thoughts that they had while choosing which brand of portable CD player to select. Finally, we gave participants the option to reverse their choice.

In Phase 2, the manipulation of type of disconfirming information occurred with a comparative advertisement praising either the rejected alternative or an alternative that was not in the initial choice set on an attribute (i.e., a car kit). We randomly assigned participants to a condition. In all experimental conditions, the advertisement showed a description of the promoted product that, except for the additional information about the car kit, was identical to the original description of the rejected alternative. To ensure that each participant received a booklet in which the brand of portable CD player he or she initially chose was attacked in the comparative advertisement, we handed booklets to the participants individually. We told them that booklets were handed to them individually because the advertisements they were to evaluate varied depending on gender and ethnic background.

After evaluating the advertisements, participants listed their thoughts and were given the chance to change their choice. Their booklets included the return slip. In an attempt to minimize the number of irrelevant thoughts, we used more specific instructions than those given in the pilot studies. Specifically, we asked participants to list the thoughts and feelings they had about "their choice of a portable CD player." The instructions also stated that the

researchers were "*not* interested in comments on the ad format." Wansink, Ray, and Batra (1994) argue for the use of "stylized" cognitive response elicitation procedures and provide guidelines for doing so without affecting the dependent variables in a study. We followed these guidelines in wording the thought-listing exercise instructions in Phase 2 of Study 1.

*The dependent variable and other measures*. We used both the expressed and behavioral measures applied in the pilot studies to assess the dependent variable (i.e., likelihood of product returns). In addition, two judges who were blind to the study hypotheses independently evaluated the cognitive responses obtained through thought listing in both phases of the study. All cognitive responses in Phase 1 of the study were classified into comparative and noncomparative thoughts. Thoughts that directly referred to a comparison between the two choice alternatives through (1) the use of relative adjectives, (2) a reference to the importance of the discrepancy in the alternatives' performance on an attribute, or (3) the questioning of the performance of both brands on an attribute that was not described were classified as comparative thoughts. All other thoughts, except idiosyncratic thoughts, were considered noncomparative. We discarded idiosyncratic thoughts from the analysis.

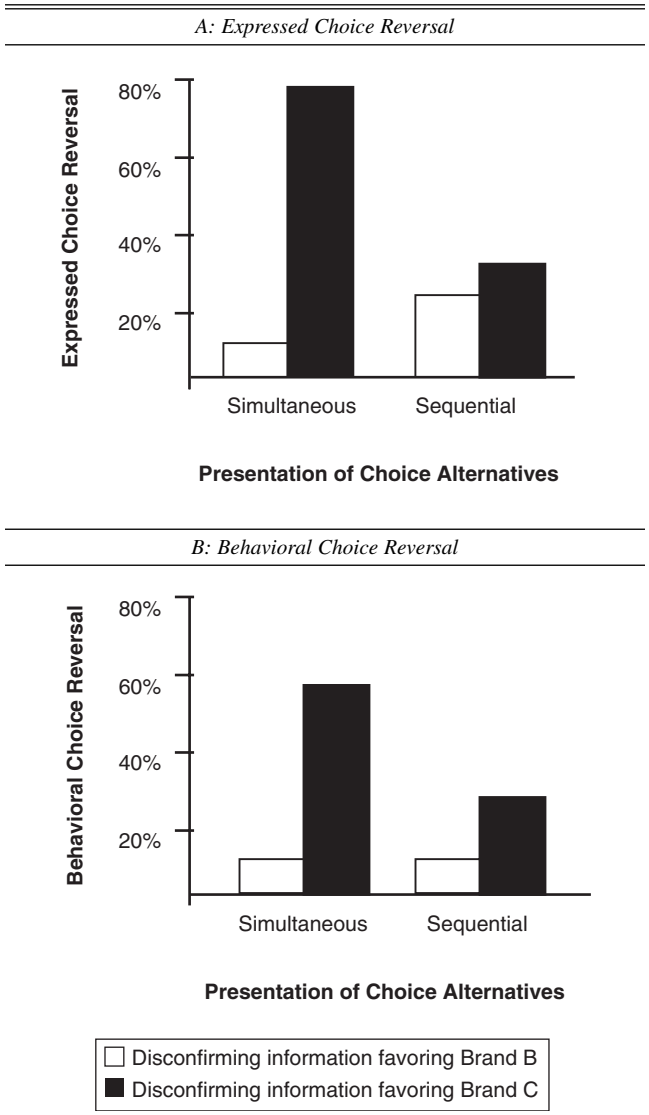
*Results*

*Product returns*. Figure 2 illustrates the percentages of participants who reversed their choice (Panel A) and those who submitted the return slip (Panel B). We logistically regressed choice reversibility that participants expressed by making a choice different from their initial one on two variables: presentation of choice sets and type of disconfirming information. We found an interaction effect (Wald [1] = 4.57, $p = .032$). Notably, the type of disconfirming information showed a strong main effect (Wald [1] = 9.22, $p = .002$), and exposure to information favoring a new alternative (Brand C) resulted in more reversibility than exposure to information favoring an already considered and rejected alternative (Brand B) (54.8% versus 25%, $\chi^2(1) = 5.55$, $p = .018$). The likelihood to return products did not differ on the basis of the mode of presentation (simultaneous: 48.28%, sequential: 33.33%; Wald [1] = 1.98, $p > .15$). An analysis of the number of participants who filled out and detached the return slip (Figure 2, Panel B) shows that the effect of the type of disconfirming information had a Wald (1) = 7.72, $p = .005$, and that the interaction effect had a Wald (1) = 4.18, $p = .04$.

Further analysis revealed that when participants were exposed to disconfirming information favoring Brand B, the percentage who reversed their decision was not affected by the mode of presentation (sequential: 28.57%, simultaneous: 21.4%; z = .14, $p > .8$), as we predicted in $H_1$. As we also posited in $H_1$, when participants were exposed to disconfirming information favoring Brand C, those in the sequential presentation condition showed a lower rate of choice reversibility (37.5%) than those in the simultaneous condition (73.3%; z = 3.36, $p < .001$).

*Cognitive responses*. All cognitive responses generated in Phase 1 of the experiment were classified into comparative and noncomparative thoughts on the basis of the coding schema we previously described. The level of agreement between the two judges exceeded 90%, and they resolved disagreements with discussion. As we expected, participants in the simultaneous condition generated more comparative thoughts ($\bar{x} = 2.25$, s = 1.61) than did those in the sequential condition ($\bar{x} = .98$, s = 1.13; t[85] = 4.29, $p < .001$). As we also expected, participants in the sequential condition generated more noncomparative thoughts ($\bar{x} = 2.43$, s = 1.25) than did those in the simultaneous condition ($\bar{x} = 1.72$, s = 1.30; t[85] = 2.61, $p = .01$). As a result, the average ratio of comparative to noncomparative thoughts among participants in the simultaneous condition ($\bar{x} = 1.54$, s = 1.55) exceeded that of participants in the sequential condition ($\bar{x} = .61$, s = .78; t[75] = 3.26, $p = .002$). The total number of thoughts generated by participants did not vary across experimental conditions, implying that effort was not driving differential choice resistance. Despite the specific instructions given for the

Figure 2
LIKELIHOOD OF PRODUCT RETURNS BY PRESENTATION OF CHOICE ALTERNATIVES AND TYPE OF DISCONFIRMING INFORMATION: EXPERIMENT 1



thought-listing exercise in Phase 2, the number of irrelevant thoughts was too overwhelming for any meaningful analysis.

## STUDY 2

### Method

*Participants and design*. Study 2 consisted of a 2 (level of negative information about the chosen alternative [Brand A] at the prepurchase stage: none and mild) × 2 (presentation of choice set: sequential and simultaneous) between-subjects design. A total of 118 undergraduate students at a northeastern university participated in the experiment for extra course credit. Their participation also enabled them to partake in a raffle with a chance to win an electronic device.

Similar to Study 1, we ran the experiment in two phases with a two-day delay. In the first phase of the study, decision makers chose between two alternative brands of a portable CD player. After two days, we exposed them to disconfirming information in the form of a comparative advertisement. In all experimental conditions, the advertisement favored a new alternative (Brand C) over the chosen brand (Brand A) by praising the existence of a car kit in Brand C. We then gave participants the option to return the product they "purchased" in the first phase of the study. In both phases, participants were seated in a behavioral lab in front of a computer. The program, which was written in MediaLab (Empirisoft 2004), provided detailed on-screen instructions.

*Manipulation of independent variables*. We provided information about the two alternatives (Brand A and Brand B) in the form of a brand-by-attribute matrix. We described each brand on four attributes: ease of use, sound accuracy, availability of low-battery indicator, and average alkaline battery life. In the no-negative-information condition, Brand A outperformed Brand B on all four attributes. In the mild-negative-information condition, Brand B outperformed Brand A on average life of an alkaline battery, an attribute that the pretests showed to be of mild importance.

We presented descriptions of the two brands either simultaneously (i.e., on the same screen) or sequentially (i.e., on two consecutive screens). We randomly assigned participants to the two presentation conditions. In both conditions, participants made their choice on a separate screen. We rotated the order of the choice alternatives (Brand A or Brand B) for the CD player.

*The dependent variable and other measures*. We used both the expressed and behavioral measures applied in Study 1 to assess the dependent variable (i.e., likelihood of product returns). For the behavioral measure of reversibility, we checked whether participants who expressed an intention to reverse their choice filled out a return slip and e-mailed it at the end of the experiment.

In addition, two judges who were blind to the study hypotheses independently evaluated the cognitive responses obtained through the thought-listing exercises in both phases of the study. The two judges used a coding schema similar to the one applied in Study 1. Thus, thoughts were categorized into comparative, noncomparative, and irrelevant thoughts. In cases in which thoughts of varying valences (i.e., in favor and not in favor of Brand A) occurred, net arguments were reported.

In an attempt to identify the thoughts retrieved from memory, the judges compared the thought listings in the two phases of the experiment, looking for cognitive responses that a participant had in the first phase of the experiment and repeated in the second phase. To minimize the number of irrelevant thoughts, we used the specific instructions given in Study 1. Participants listed each thought on a separate screen and then hit the "Enter" key to move to the next screen. We measured the time period between hitting the Enter key and hitting the first key when typing on the thought-listing screen in milliseconds. Previous researchers have used response latency as an indicator of accessibility (see, e.g., Fazio, Powell, and Williams 1989).

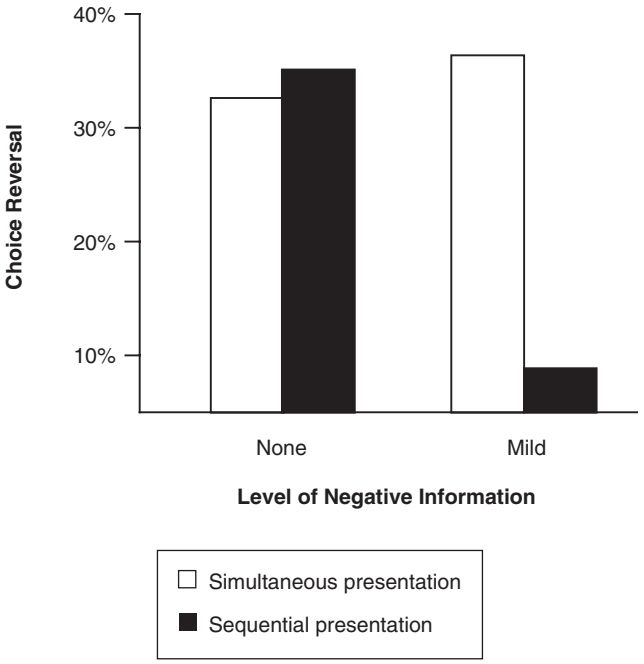
### Results

*Manipulation checks*. To use the data in this study, consumers across condition must choose Brand A in the first phase of the study. Thus, we conducted considerable pretesting to ensure the before-attack superiority of Brand A. In the actual study, however, 17 participants preferred Brand B to Brand A. We discarded these participants' data from the analysis, which dropped the number of participants to 101.

We also ran a 2 (presentation of choice set) × 2 (negative information) analysis of variance (ANOVA) on the importance of the availability of a car kit in a CD player, which revealed no significant effects. This finding indicates that participants across experimental conditions perceived the postpurchase attack on their choice as equally strong.

*Main results*. Figure 3 illustrates the percentages of participants who reversed their choice (i.e., made a final choice

Figure 3
LIKELIHOOD OF PRODUCT RETURNS BY PRESENTATION OF CHOICE ALTERNATIVES AND LEVEL OF NEGATIVE INFORMATION: EXPERIMENT 2



[Brand C] that differed from their initial one [Brand A]) in each of the experimental conditions. Given the categorical nature of the dependent variable (i.e., participants either reversed their decision or did not), we ran a logistic regression with presentation and level of negative information as independent variables. The analysis revealed an interaction effect (Wald [1] = 3.89, $p < .05$). Note that all 28 participants who made a final choice that was different from their initial one filled out and e-mailed the return slip to activate their choice reversibility.

In high-involvement choice situations, $H_{2a}$ predicts that when inoculated consumers are faced with disconfirming information favoring Brand C, those who are presented with choice alternatives sequentially are less likely to return Brand A (7.4%) than consumers who are exposed solely to positive information about Brand A at that same prepurchase stage (34.6%; z = 2.38, $p < .05$). Our results support this hypothesis. As we predict in $H_{2b}$, inoculated consumers who were presented with choice alternatives sequentially were less likely to return Brand A (7.4%) than inoculated consumers in the simultaneous presentation condition (37%; z = 2.10, $p < .05$).

In Phase 1, to code participants' thoughts, the two judges applied the coding schema described in Study 1. The level of agreement between the two judges was 89%. They resolved disagreements with discussion. We ran a 2 (presentation) × 2 (negative information) ANOVA on the number of total relevant thoughts, which revealed a main effect for level of negative information ($F(1, 97) = 57.02$, $p < .001$). As we expected, inoculated participants generated more thoughts ($\bar{x} = 2.70$) than participants who were exposed to only positive information ($\bar{x} = 1.38$; t[99] = 7.54, $p < .001$). Similar to Study 1, participants in the simultaneous presentation condition listed more comparative thoughts than did participants in the sequential presentation condition ($\bar{x} = 1.33$ versus $\bar{x} = .62$; $p < .05$). Sequential presentation led to more noncomparative thoughts than simultaneous presentation ($\bar{x} = 1.52$ versus $\bar{x} = .63$; $p < .05$).

In Phase 2, the judges compared thought listings in the two phases of the experiment and found that 18 of 101 participants repeated at least one thought that they had listed in Phase 1 of the experiment. Table 1 shows the percentages of participants who stated previously listed thoughts. Participants' likelihood to reverse their choice was negatively related to participants' listing of previously recorded thoughts ($\chi^2(1) = 6.807$, $p < .01$).

Recalled thoughts were listed more quickly than were nonrecalled ones (3207 versus 2691, t[99] = 3.65, $p = .001$). Moreover, we ran an ANOVA on latency with presentation, negative information, and their interaction as independent variables, which resulted in an interaction effect ($F(1, 97) = 4.60$, $p = .034$). Table 1 shows mean latencies in the various experimental conditions.

Table 1
RECALL AND LATENCY: EXPERIMENT 2

| *Presentation* | *Negative* | *Percentage of Participants Recalling Thoughts* | *Mean Latencies (in Milliseconds)* |
|---|---|---|---|
| Simultaneous | None | 4.8 | 3117 |
| | Mild | 11.1 | 3235 |
| Sequential | None | 7.7 | 3222 |
| | Mild | 44.4 | 2890 |

## *DISCUSSION*

In both Studies 1 and 2, buying processes leading to the generation of different thoughts at the prechoice stage resulted in a differential likelihood of product returns. Those findings provide support for a clear link between what happens before choice and what happens after purchase. In Study 1, the comparative versus noncomparative nature of thoughts generated influenced their diagnosticity and, accordingly, their impact on product returns when consumers were exposed to disconfirming information. Diagnosticity was also a function of the type of attack the consumers faced (i.e., rejected brand versus new brand).

The findings from Study 1 revealed a main effect, which was not hypothesized initially, for the type of disconfirming information, showing that decision reversibility was greater when participants were given the option to choose an alternative that was not initially in the choice set than when they were given the option to choose a rejected alternative. This main effect implies that either decision makers find it difficult to admit their mistake (Staw 1976) or newly encountered alternatives have a novelty advantage. Another possible explanation can be derived from previous research that shows that information on the previously chosen brand is highly accessible and that participants try to retrieve the chosen brand's information (Biehal and Chakravarti 1986). The same is not necessarily true for a rejected brand (Brand B in this context). Thus, information about Brand B is not retrieved, and therefore we expect less choice reversal than when a new brand (Brand C) is added.

The findings of this research extend inoculation theory and help delineate some of its boundaries. Study 2 provides support for the role of inoculation in choice resistance. Previous researchers have found that inoculation results in more resistant attitudes (Godbold and Pfau 2000). This research extends the use of inoculation to include choice situations.

Although our studies provide support for the impact of the prechoice buying process on postpurchase behavior, the mechanism underlying this impact remains unclear. The overwhelming occurrence of irrelevant thoughts in Phase 2 of Study 1 made it impossible to capture any retrieval process that might have occurred. Other researchers also refer to the difficulty of eliciting certain internal processes using thought listing (Bettman, Johnson, and Payne 1991). In Study 2, repeated thoughts occurred, and their latency was shorter. However, although some researchers use latency as an indicator of accessibility (Fazio, Powell, and Williams 1989), other researchers link it to certainty of the answer (MacLachlan, Czepiel, and LaBarbera 1979). Moreover, other researchers note the difficulty of separating an information accessibility explanation from alternative explanations in mixed-choice situations (Biehal and Chakravarti 1986). However, if thoughts were not retrieved and were indeed generated in Phase 2, it is not clear why they would be generated more quickly than other new thoughts. Our latency results suggest that part of the impact of prechoice cognitive responses is unconscious (i.e., below awareness levels).

The major contribution of this research is its emphasis on the value of examining the nature of cognitive responses to understand underlying aspects of the choice process. We examined this along two dimensions. First, by coding thoughts as net arguments in favor of the chosen alternative (i.e., by deducting the number of nonsupportive arguments from the supportive ones), we accounted for the possibility that decision makers have conflicting thoughts about an issue or a product, which is strongly supported by many researchers (Bettman et al. 1993). Second, we took a holistic (global) approach to choice in which we classified cognitive responses that consumers develop in choice situations into comparative and noncomparative thoughts. This classification is a simplification of the far more detailed one that Bettman and Park (1980) apply to coding verbal protocols. The results indicate that this simplified version can help uncover some of the decision techniques governing choice.

### *LIMITATIONS*

Our findings should be interpreted with caution for several reasons. First, the reliance on retrospective thought listing to understand the prechoice process has several limitations (for a detailed discussion, see Ericsson and Simon 1999). Second, the classification of cognitive responses as comparative and noncomparative needs further refinement. Third, of the various reasons to return products, the focus of this research is on returns resulting from exposure to new, disconfirming information. Given that diagnosticity of thoughts is a key factor in our model and that diagnosticity varies with the type of the attack (e.g., whether it brings in new information, how specific or general it is), examining whether the major premise in our model holds under different forms of attack is necessary to test the versatility of the model. Fourth, in our studies, Brands B and C are similar. Having rejected Brand B, participants might be hesitant to reverse their choice in favor of a similar alternative, Brand C. Thus, it is necessary to study choice reversibility while manipulating the extent to which alternatives are similar.

### *IMPLICATIONS*

This research approaches product returns differently from previous research. As a result, it has implications for managers. Previous researchers have investigated attempts to reduce returns through, for example, the adoption of strict return policies (Wood 2001) or through trying to save sales by helping consumers when they come to the store to return products. Other research has focused on what needs to be done to reduce the costs of handling returns, such as redesigning the logistics of return. In this research, the implication is that managers might be able to affect buyers' likelihood of product returns through cleverly designing the presentation of alternatives.

The results indicate that comparative presentations, though effective against previously encountered brands, are vulnerable to disconfirming evidence favoring new brands. Thus, manufacturers may be better off placing their products in an arrangement that discourages simultaneous comparison. This is particularly true in industries in which new brands are introduced often. Some manufacturers, such as Benetton and Gap, have their own stores to sell their products. Other manufacturers do not place their products in stores that carry competing products. Our findings encourage similar behavior.

Sequential and simultaneous presentation formats are not manipulated only through store choice or design. Other nonstore contexts may also affect the ease or difficulty of comparison formats. Some Web sites are designed intentionally to facilitate simultaneous comparisons. Designs of trade shows and formats in which individual offers are presented to customers could encourage or discourage some comparison formats. Managers concerned about product returns may want to consider the resulting comparison format when deciding on outlets for their products.

The results also raise questions about the efficacy of some of the push strategies and direct response techniques that many marketing managers apply. The results indicate that strategies that do not encourage consumers to generate as many cognitive responses in favor of the chosen alternative as other strategies often result in relatively high product return rates. Marketers often announce special offers for limited periods of time, such as calling within the next few minutes or being one of the first few callers. Our findings suggest that this approach results in high product returns because under these circumstances, customers have a little opportunity to generate cognitive responses in favor of the chosen alternative (Wright 1974).

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